Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The inflection points on the graph of function $y = \int_0^x (t-1)(t-2)^2 \, \mathrm{d}t$ are

(A) $x = -1$

(B) $x = 3/2$

(C) $x = 4/3$

(D) $x = 1$

Q.2

The value of the integral $\int_0^{100\pi} \sqrt{1 - \cos 2x} \, \mathrm{d}x$ is

(A) $100\sqrt{2}$

(B) $200\sqrt{2}$

(C) 0

(D) 100π

Q.3

Let f be a periodic continuous function with period $T > 0$. If $I = \int_0^T f(x)dx$ Then the value of $I_1 = \int_4^{4+4T} f(3x)dx$ is

(A) I

(B) $2I$

(C) $3I$

(D) $4I$

Q.4

If $g(x) = \int_0^x \cos 4t dt$, then $g(x+\pi)$ equals

(A) $g(x)/g(\pi)$

(B) $g(x) - g(\pi)$

(C) $-g(x)g(\pi)$

(D) $g(x)g(\pi)$

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Q.5

If $f(x) = \int_0^x \sin^8 t \, dt$, then $f(x+\pi)$ equals

(A) $\frac{f(x)}{f(\pi)}$

(B) $f(x)f(\pi)$

(C) $f(x) + f(\pi)$

(D) $f(x) - f(\pi)$

Q.6

If the value of $\int_{-2}^{2} |x \cos \pi x| dx = k/\pi$ then the value of k is

(A) 4

(B) 8

(C) 12

(D) none of these

Q.7

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$, n is a positive integer. Then

(A) $I_n : I_{n-2} = n : (n-1)$

(B) $I_n > I_{n-2}$

(C) $n(I_{n-2} - I_n) = I_{n-2}$

(D) None of these

Q.8

If $u_n = \int_0^{\pi/4} \tan^n \theta d\theta$ then $u_n + u_{n-2}$ is

(A) $\frac{1}{n-1}$

(B) $\frac{1}{n+1}$

(C) $\frac{1}{2n-1}$

(D) $\frac{1}{2n+1}$

Q.9

If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$ then what is $I_n + I_{n-2}$
equal to?

(A) $\frac{1}{n}$

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(B) $\frac{1}{(n-1)}$

(C) $\frac{n}{(n-1)}$

(D) $\frac{1}{(n-2)}$

Q.10

If $f(x) = \int_{2x}^{\sin x} \cos\left(t^3\right) dt$, then $f'(x)$ is equal to

(A) $\cos\left(\sin^3 x\right)\cos x - 2\cos\left(8x^3\right)$

(B) $\sin\left(\sin^3 x\right)\sin x - 2\sin\left(8x^3\right)$

(C) $\cos\left(\cos^3 x\right)\cos x - 2\cos\left(x^3\right)$

(D) $\cos\left(\sin^3 x\right) - \cos\left(8x^3\right)$

Q.11

The line $y = \alpha$ intersects the curve $y = g(x)$, atleast at two points. If $\int_2^x g(t)dt = \frac{x^2}{2} + \int_x^2 t^2\, g(t)dt$ then possible value of α is / are $-$

(A) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

(B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

(D) $\left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}$

Q.12

Let $fy = \int_0^1 e^{x\text{-}y^2} x^3 \text{-} 2xy + 3x^2 \text{-} 2y dx$, then $\int_0^1 e^y f y dy$ is equal to

(A) 1

(B) $e - 1$

(C) 0

(D) $e^2 - 2e$

Q.13

Let $f : R \to R$ is differentiable function and $f(1) = 4$, then the value of $\lim_{x\to 1} \int_0^{f(x)} \frac{2tdt}{x-1}$ is

(A) $8f'(1)$

(B) $4f'(1)$

(C) $2f'(1)$

(D) $f'(1)$

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Q.14

The line $y = \alpha$ intersects the curve $y = g(x)$, atleast at two points. If $\int_2^x g(t)dt = \frac{x^2}{2} + \int_x^2 t^2\, g(t)dt$ then possible value of α is/are $-$

(A) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

(B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

(D) $\left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}$

Q.15

The function $f(x) = \int_{-1}^{x} t\,(e^t - 1)\,(t-1)(t-2)^3(t-3)^5$ dt has a local minimum at $x =$

(A) 0

(B) 1,3

(C) 2

(D) None of these

Q.16

The tangent of the curve $y = f(x)$ at the point with abscissa $x = 1$ form an angle of $\pi/6$ and at the point $x = 2$ an angle of $\pi/3$ and at the point $x = 3$ an angle of $\pi/4$. If $f'(x)$ is continuous, then the value of $\int_1^3 f''(x)f'(x)dx + \int_2^3 f''(x)dx$ is

(A) $\frac{4\sqrt{3}-1}{3\sqrt{3}}$

(B) $\frac{3\sqrt{3}-1}{2}$

(C) $\frac{4-3\sqrt{3}}{3}$

(D) none of these

Q.17

Let $f : (0, \infty) \to R$ and $F(x) = \int_0^x f(t)dt$
If $F\left(x^2\right) = x^2(1+x)$, then $f(4)$ equals

(A) $\frac{5}{4}$

(B) 7

(C) 4

(D) 2

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Q.18

Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t} dt$
Then $F(e)$ equals

(A) 1

(B) 2

(C) $1/2$

(D) 0

Q.19

Find the value of $\int_0^9 [\sqrt{x} + 2] dx$ where $[.]$ is the greatest integer function :

(A) 31

(B) 22

(C) 23

(D) None of these

Q.20

What is $\int_0^{\pi/2} \sin 2x \ell n(\cot x) dx$ equal to ?

(A) 0

(B) $\pi \ell n 2$

(C) $-\pi \ln 2$

(D) $\frac{\pi \ell n 2}{2}$

Answers & Solutions

Q.1 Answer:

$x = 4/3$

Solution:

$\frac{dy}{dx} = (x-1)(x-2)^2$ **so** $\frac{d^2y}{dx^2} = (x-2)(3x-$

4). The points of inflection are given by $\frac{d^2y}{dx^2} = 0$ **so** $x = 2, x = 4/3$ **are points of inflection.**

Q.2 Answer:

$200\sqrt{2}$

Solution:

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We have $\sqrt{1-\cos 2x}=\sqrt{2}|\sin x|$. Since
the period of $|\sin x|$ is π, so $\int_0^{100\pi}\sqrt{1-\cos 2x}\,\mathrm{d}\,x=\sqrt{2}\int_0^{100\pi}|\sin x|\mathrm{d}x=$
$100\sqrt{2}\int_0^{\pi}\sin x\,\mathrm{d}x=200\sqrt{2}$.

Q.3 Answer:

$4I$

Solution:

Put $3x=y$ in I_1
$I_1=\frac{1}{3}\int_{12}^{12+12T}f(y)dy$
$=\frac{1}{3}\left[\int_{12}^{T}f(y)dy+\sum_{K=1}^{11}\int_{KT}^{(K+1)T}f(y)dy+\int_{12T}^{12+12T}f(y)dy\right]$
But $\int_{KT}^{(K+1)T}f(y)dy=\int_0^T f(u)du$,
(Put $KT+u=y$ or use Property 17) and
$\int_{12T}^{12+12T}f(y)dy=\int_0^{12}f(u)du$
Hence

$$I_1=\frac{1}{3}\left[\int_{12}^{T}f(y)dy+11I+\int_0^{12}f(u)du\right]$$
$$=\frac{1}{3}\left[\int_0^T f(y)dy+11I\right]=\frac{1}{3}\times 12I=4I.$$

Q.4 Answer:

$g(x)-g(\pi)$

Solution:

$g(x+\pi)=\int_0^{x+\pi}\cos 4tdt$
$=\int_0^{\pi}\cos 4tdt+\int_{\pi}^{x+\pi}\cos 4tdt$
$=g(\pi)+I$
where $I=\int_{\pi}^{x+\pi}\cos 4tdt$, Put $t=\pi+\theta$, so that

$$I=\int_0^x\cos(4\pi+4\theta)d\theta=\int_0^x\cos 4\theta d\theta=g(x)$$

So $g(x+\pi)=g(\pi)+g(x)$ but
$\therefore\quad g(x+\pi)=g(x)-g(\pi)$ also.

Q.5 Answer:

$f(x)+f(\pi)$

Solution:

$f(x+\pi)=\int_0^{x+\pi}\sin^8 tdt$
$=\int_0^{\pi}\sin^8 tdtt\int_{\pi}^{x+\pi}\sin^8 tdt$

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$= f(x) + \int_0^x \sin^8(\pi - u)du = f(\pi) + f(x).$

Q.6 Answer:

8

Solution:

Since the integrand is an even function so required integral is equal to $2\int_0^2 |x\cos\pi x|dx$

$= 2\left[\int_0^{1/2} x\cos\pi x dx - \int_{1/2}^{3/2} x\cos\pi x dx + \int_{3/2}^2 x\cos\pi x dx\right]$

$= 2\left[x\frac{\sin\pi x}{\pi} + \frac{\cos\pi x}{\pi^2}\Big|_0^{1/2} - \left(x\frac{\sin\pi x}{\pi} + \frac{\cos\pi x}{\pi^2}\right)\Big|_{1/2}^{3/2}\right.$

$\left. + \left(\frac{x\sin\pi x}{\pi} + \frac{\cos\pi x}{\pi^2}\right)\Big|_{3/2}^2\right]$

$= 2\left[\frac{1}{2\pi} - \frac{1}{\pi^2} - \left(-\frac{3}{2\pi} - \frac{1}{2\pi}\right) + \frac{1}{\pi^2} - \left(-\frac{3}{2\pi}\right)\right] = \frac{8}{\pi}$

Q.7 Answer:

$\mathrm{n}\left(\mathrm{I}_{\mathrm{n}-2} - \mathrm{I}_{\mathrm{n}}\right) = \mathrm{I}_{\mathrm{n}-2}$

Solution:

$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \sin^{n-1} x \sin x dx$

$\left[-\sin^{n-1} x\cos x\right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1)\sin^{n-2} x\cos^2 x dx$

$= (\mathrm{n}-1)\int_0^{\frac{\pi}{2}} \sin^{\mathrm{n}-2}\mathrm{x}\left(1-\sin^2\mathrm{x}\right)\mathrm{dx}$

$= (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1)\int_0^{\frac{\pi}{2}} \sin^n x dx$

$\therefore \quad I_n = (n-1)I_{n-2} - (n-1)I_n$

$\Rightarrow \mathrm{n}\mathrm{I}_{\mathrm{n}} = (\mathrm{n}-1)\mathrm{I}_{\mathrm{n}-2} \Rightarrow \mathrm{I}_{\mathrm{n}} = \frac{\mathrm{n}-1}{\mathrm{n}}\mathrm{I}_{\mathrm{n}-2}$

$\Rightarrow \mathrm{n}\left(\mathrm{I}_{\mathrm{n}-2} - \mathrm{I}_{\mathrm{n}}\right) = \mathrm{I}_{\mathrm{n}-2}$

Also, $I_n : I_{n-2} = (n-1) : n$ **and** $I_{n-2} > I_n$

Q.8 Answer:

$\frac{1}{n-1}$

Solution:

Given: $\mathrm{u}_{\mathrm{n}} = \int_0^{\pi/4} \tan^{\mathrm{n}}\theta \mathrm{d}\theta$

$= \int_0^{\pi/4} \tan^2\theta \tan^{\mathrm{n}-2}\theta \mathrm{d}\theta$

$= \int_0^{\pi/4} \sec^2\theta \tan^{\mathrm{n}-2}\theta \mathrm{d}\theta - \int_0^{\pi/4} \tan^{\mathrm{n}-2}\theta \mathrm{d}\theta$

$= \int_0^{\pi/4} \sec^2\theta \tan^{\mathrm{n}-2}\theta \mathrm{d}\theta - \mathrm{u}_{\mathrm{n}-2}$

$= \frac{\tan^{\mathrm{n}-1}\theta}{\mathrm{n}-1}\Big|_0^{\pi/4} = \frac{1}{\mathrm{n}-1}$

Q.9 Answer:

$\frac{1}{(n-1)}$

Solution:

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Let $I_n = \int_0^{\pi/4} \tan^n x dx$

Consider,

$I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x dx + \int_0^{\pi/4} \tan^{n-2} x dx$

$= \int_0^{\pi/4} \tan^{n-2} x \left(\tan^2 x + 1\right) dx$

$= \int_0^{\pi/4} \sec^2 x \tan^{n-2} x dx$

$\text{Put} \tan x = t$

$\sec^2 x dx = dt$

when $x = 0$ **then** $t = 0$ **and when** $x = \frac{\pi}{4}, t = 1$

$\therefore I_n + I_{n-2} = \int_0^1 t^{n-2} dt$

$= \frac{t^{n-2+1}}{n-2+1}\Big|_0^1 = \frac{t^{n-1}}{n-1}\Big|_0^1 = \frac{1}{n-1}[1-0] = \frac{1}{n-1}$

Q.10 Answer:

$\cos\left(\sin^3 x\right)\cos x - 2\cos\left(8x^3\right)$

Solution:

$f(x) = \int_{2x}^{\sin x} \cos\left(t^3\right) dt; f'(x) = \frac{d}{dx}\int_{2x}^{\sin x} \cos\left(t^3\right) dt$

$= f(\sin x) \cdot \cos x - f(2x) \times 2$

$= \cos\left(\sin^3 x\right) \cdot \cos x - \cos(2x)^3 \times 2$

$= \cos\left(\sin^3 x\right) \cdot \cos x - 2\cos\left(8x^3\right)$

Q.11 Answer:

$\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

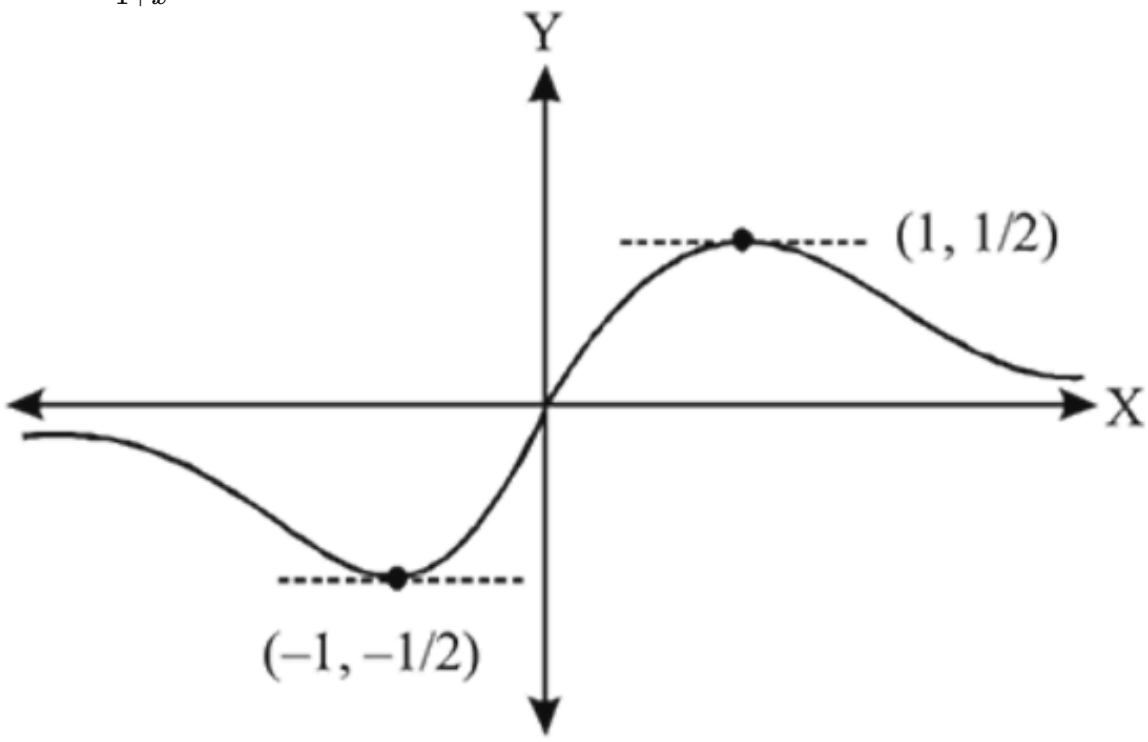
Solution:

$\int_2^x g(t)dt = \frac{x^2}{2} + \int_x^2 t^2 g(t)dt$

Differentiating w.r.t. x, we get $g(x) = x + \left(-x^2(g(x))\right.$

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$\Rightarrow g(x) = \frac{x}{1+x^2}$



Clearly from graph, $\alpha \in \left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

Q.12 Answer:

0

Solution:

$fy = e^{-y^2} \int_0^1 e^x x^3 - 2xy + 3x^2 - 2y dx$

$= e^{-y^2} e^x x^3 - 2xy \Big|_0^1 = e^{-y^2}\left(e1 - 2y - 0\right) = e^{1-y^2} 1 - 2y$

$\therefore \int_0^1 e^y fy dy = \int_0^1 e^{y+1-y^2} 1 - 2y dy$

$= \int_1^1 e^t dt = 0$

Q.13 Answer:

$8f'(1)$

Solution:

$\lim_{x\to 1} \int_0^{f(x)} \frac{2t}{x-1} dt$

$= \lim_{x\to 1} \frac{1}{x-1} \left[t^2\right]^{f(x)} = \lim_{x\to 1} \frac{f^2(x) - f^2(0)}{x-1}$

$= \lim_{x\to 1} \frac{2f(x)f'(x)}{1}$ **(L'Hospital Rule)**

$= 2f(1)f'(1) = 8f'(1)$

Q.14 Answer:

$\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

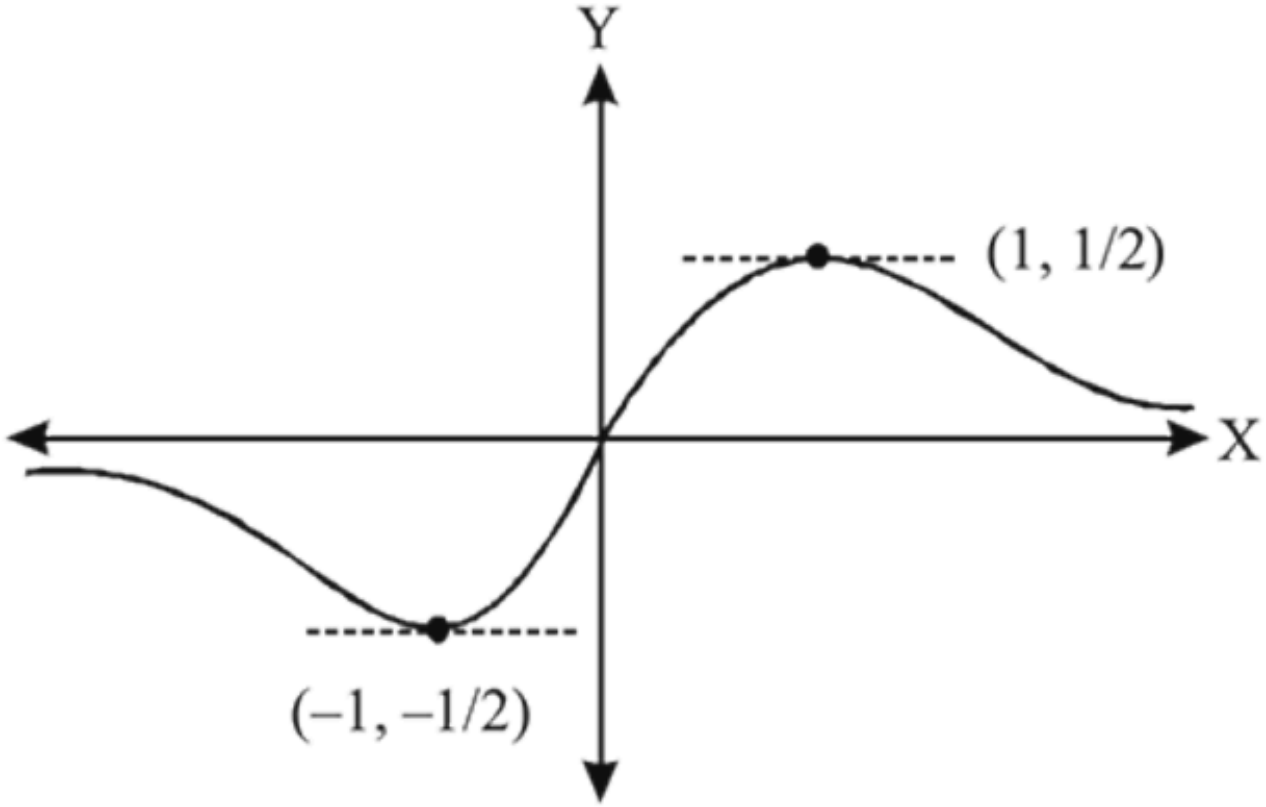
Solution:

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$\int_2^x g(t)dt = \frac{x^2}{2} + \int_x^2 t^2 g(t)dt$

Differentiating w.r.t. x, we get

$g(x) = x + \left(-x^2(g(x)) \Rightarrow g(x) = \frac{x}{1+x^2}\right.$



Clearly from graph, $\alpha \in \left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

Q.15 Answer:

1,3

Solution:

$\frac{dy}{dx} = f'(x)$

$\Rightarrow x\,(e^x - 1)\,(x-1)(x-2)^3(x-3)^5 = 0$

Critical points are $0, 1, 2, 3$ **Consider change of sign of** $\frac{\mathrm{dy}}{\mathrm{dx}}$ **at** $\mathrm{x} = 3.$

$x < 3$, $\frac{dy}{dx}$ **= negative and** $x > 3$, $\frac{dy}{dx}$ **= positive**

Change is from negative to positive, hence minimum at $\mathrm{x} = 3.$ **Again minimum and maximum occur alternately.**

. 2nd minimum is at $\mathrm{x} = 1.$

Q.16 Answer:

$\frac{4-3\sqrt{3}}{3}$

Solution:

According to question $\mathrm{f}'(1) = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$

$\mathrm{f}'(2) = \tan\frac{\pi}{3} = \sqrt{3}$ **and** $\mathrm{f}'(3) = \tan\frac{\pi}{4} = 1$

so, $\int_1^3 f''(x)f'(x)dx + \int_1^3 f''(x)dx$

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$= \left[\frac{\{f'(x)\}^2}{2}\right]_1^3 + [f'(x)]_2^3$
$= \frac{1}{2}\left[1 - \frac{1}{3}\right] + [1 - \sqrt{3}] = \frac{4}{3} - \sqrt{3}$

Q.17 Answer:

4

Solution:

$F'(x) = f(x)$ **Also,** $F(t) = t(1 + \sqrt{t})$
$\Rightarrow \quad F'(t) = 1 + \frac{3}{2}t^{1/2}; F'(4) = 1 + 3 = 4 \Rightarrow f(4) = 4$

Q.18 Answer:

$1/2$

Solution:

Given $F(x) = f(x) + f\left(\frac{1}{x}\right)$**, where**
$f(x) = \int_1^x \frac{\log t}{1+t} dt$
$F(e) = f(e) + f\left(\frac{1}{e}\right)$
$\Rightarrow F(e) = \int_1^e \frac{\log t}{1+t} dt + \int_1^{1/e} \frac{\log t}{1+t} dt \ldots . (\mathrm{A})$
Now for solving, $I = \int_1^{1/e} \frac{\log t}{1+t} dt$
Put $\frac{1}{t} = z \Rightarrow -\frac{1}{t^2} dt = dz \Rightarrow dt = -\frac{dz}{z^2}$
and limit for $t = 1 \Rightarrow z = 1$ **and for** $t = 1/e \Rightarrow z = e$
$\therefore \quad I = \int_1^e \frac{\log\left(\frac{1}{z}\right)}{1+\frac{1}{z}} \left(-\frac{dz}{z^2}\right)$
$= \int_1^e \frac{(\log 1 - \log z) \cdot z}{z+1} \left(-\frac{dz}{z^2}\right)$
$= \int_1^e -\frac{\log z}{(z+1)} \left(-\frac{dz}{z}\right) = \int_1^e \frac{\log z}{z(z+1)} dz$
$\therefore I = \int_1^e \frac{\log t}{t(t+1)} dt$
Equation becomes:
$F(e) = \int_1^e \frac{\log t}{1+t} dt + \int_1^e \frac{\log t}{t(1+t)} dt$
$= \int_1^e \frac{t \cdot \log t + \log t}{t(1+t)} dt = \int_1^e \frac{(\log t)(t+1)}{t(1+t)} dt$
$\Rightarrow F(e) = \int_1^e \frac{\log t}{t} dt$
Let $\log t = x \quad \therefore \frac{1}{t} dt = dx$
[**for limit** $t = 1, x = 0$ **and** $t = e, x = \log e = 1$]
$\therefore F(e) = \int_0^1 x dx \quad F(e) = \left[\frac{x^2}{2}\right]_0^1 \Rightarrow F(e) = \frac{1}{2}$

Q.19 Answer:

31

Solution:

We have $\int_0^9 [\sqrt{x} + 2] dx$
$= \int_0^1 2 \, \mathrm{dx} + \int_1^4 3 \, \mathrm{dx} + \int_4^9 4 \, \mathrm{dx}$

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$= 2 + (12 - 3) + (36 - 16) = 2 + 9 + 20 = 31$

Q.20 Answer:

0

Solution:

Let $I = \int_0^{\frac{\pi}{2}} \sin 2x \ell n(\cot x) dx$
$= \int_0^{\frac{\pi}{2}} \sin 2x \ln(\cos x) dx - \int_0^{\frac{\pi}{2}} \sin 2x \ln(\sin x) dx$
$= \int_0^{\frac{\pi}{2}} \sin 2x \ln(\sin x) dx$
$= \int_0^{\frac{\pi}{2}} \sin(\pi + 2x) \ln(\sin x) dx$
$= \int_0^{\frac{\pi}{2}} \sin 2x \ln(\sin x) dx - \int_0^{\frac{\pi}{2}} \sin 2x \ln(\sin x) dx$
$= 0$

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